

## 11.04 Graphing Rational Functions

Common problems and solution methods

### Rule guide: recognize the task first

| Recognize                                                    | Apply these rules                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Functions And Models                                         | Read the function form and the context before choosing a graphing or modeling method.                          |
| The expression contains the chapter's key notation or model. | Write the known quantities, choose the matching formula, calculate carefully, and check the result in context. |

### Example 1: Identify the model

#### Problem

$$f(x) = x^2$$

#### How to solve it

Write the known quantities and restrictions.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Check the result against the original statement.

### Example 2: Apply the key rule

#### Problem

$$y = 2(x - 3)^2 - 5$$

#### How to solve it

Write the known quantities and restrictions.

$$\text{vertex of } y = a(x - h)^2 + k : (h, k)$$

Check the result against the original statement.

### Example 3: Check the result

#### Problem

$$\frac{f(4) - f(1)}{4 - 1}$$

#### How to solve it

Write the known quantities and restrictions.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

Check the result against the original statement.